

Name: _____

Date: _____



Year 8 Mathematics 2015

Test 6: General and Extension

Topics – Solving Equations

40

= _____ %

Total Time: **40** minutes

Total Reading: **2** minutes

Total Working: **32** minutes

Weighting: **% of the year.**

CALCULATOR ALLOWED

Time: *minutes*

Reading: *2 minutes*

Working: *40 minutes*

Total Marks : 40

Equipment Allowed: *1 A4 page of notes*

[6 marks, 1, 1, 1, 1, 1, 1]

1. Determine the value of the symbol in each of the following;

a) $\square + 19 = 21$

$\square = 2$ ✓

b) $13 - \triangle = 8$

$\triangle = 5$ ✓

c) $28 + \heartsuit = 82$

$\heartsuit = 54$ ✓

d) $\bigcirc - (-2) = 11$

$\bigcirc = 9$ ✓

e) $8 \times \star = 48$

$\star = 6$ ✓

f) $\pentagon \div 12 = 5$

$\pentagon = 60$ ✓

[6 marks, 1, 1, 1, 1, 1]

2. Determine the value of the variable (letter) in each of the following:

a) $x + 12 = 28$ $x = 16$ ✓

b) $5 - r = -4$ $r = 9$ ✓

c) $3m = 24$ $m = 8$ ✓

d) $\frac{w}{7} = 42$ $w = 294$ ✓

e) $x^3 = 8$ $x = 2$ ✓

f) $3^x = 9$ $x = 2$ ✓

[3 marks, 1, 1, 1]

3. Write an equation for the following;

a) I think of a number (x) then subtract 9 and the result is 6

$$x - 9 = 6$$
 ✓

b) I think of a number (z) then add 12 and the result is 8

$$z + 12 = 8$$
 ✓

c) I think of a number (y) double it and the result is 22.

$$2y = 22$$
 ✓

[6 marks, 1,2,1,2]

4. Solve the following equation;

a) $8d + 2 = 26$

$d = 3$ ✓

b) $\frac{7y+1}{2} = 32$

$y = 9$ ✓

$7y + 1 = 64$

$7y = 63$ ✓

c) $\frac{w}{6} - 8 = -2$

$w = 36$ ✓

$\frac{w}{6} = 6$ ✓

d) $3(3z + 4) = 40$

$z = 4$ ✓

[1 mark]

5. To hire a mower the cost is \$7 plus \$2 for every day of use. Circle the correct equation if the total charge is \$21?

a) $7d + 2 = 21$

b) $d + 7 = 21$

c) $7 + 2d = 21$ ✓

d) $d + 2 = 21$

[2 marks]

6. Brandon has 6 more marbles than Charvelle. Together they have 28 marbles. If Charvelle has h marbles, write an equation and find out how many marbles Brandon has.

$2h + 6 = 28$ ✓

Brandon = 17 marbles ✓

[4 marks, 2, 2]

7. Matthew is three times the age of Teagan plus 4 years. Matthew is 40 years old.

a) Write an equation using t to represent Teagan's age.

$$3t + 4 = 40$$

b) solve the equation to find Teagan's age.

$$3t = 36$$

$$t = 12$$

Teagan is 12 years old.

[6 marks, 1, 1, 1, 1, 1, 1]

8. For each equation determine by substitution whether the given solution in the brackets are True or False.

a) $x + 8 = 6$ [$x = -3$] F

b) $y - 4 = 4$ [$y = 8$] T

c) $5k - 31 = -1$ [$k = 6$] T

d) $\frac{3p+4}{2} = 12$ [$p = 3$] F

e) $2(d + 5) = 28$ [$d = 4$] F

f) $\frac{84}{z} - 12 = -5$ [$z = 12$] T

[6 marks, 1, 1, 2, 2]

9. Solve the following equations;

a) $6x = 5x + 17$

$$x = 17$$

b) $16q - 102 = 14q - 76$

$$q = 13$$

c) $3(z + 8) = 2(4z - 3)$

$$3z + 24 = 8z - 6$$

$$5z = 30$$

$$z = 6$$

d) $\frac{x+5}{3} = \frac{2x+7}{9}$

$$\frac{3x+15}{9} = \frac{2x+7}{9}$$

$$x = -8$$

END OF TEST